Cloud Computing Lab 4

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Screenshot 1:

```
asmithapes2ug23cs102@asmithapes2ug23cs102-VirtualBox:~$ export SRN=PES2UG23CS102
asmithapes2ug23cs102@asmithapes2ug23cs102-VirtualBox:~$ echo $SRN
PES2UG23CS102
asmithapes2ug23cs102@asmithapes2ug23cs102-VirtualBox:~$ sudo docker run hello-world
[sudo] password for asmithapes2ug23cs102:

Hello from Docker!
This message shows that your installation appears to be working correctly.

To generate this message, Docker took the following steps:
 1. The Docker client contacted the Docker daemon.
 2. The Docker daemon pulled the "hello-world" image from the Docker Hub.
    (amd64)
 3. The Docker daemon created a new container from that image which runs the
    executable that produces the output you are currently reading.
 4. The Docker daemon streamed that output to the Docker client, which sent it
    to your terminal.

To try something more ambitious, you can run an Ubuntu container with:
 $ docker run -it ubuntu bash

Share images, automate workflows, and more with a free Docker ID:
 https://hub.docker.com/

For more examples and ideas, visit:
 https://docs.docker.com/get-started/

asmithapes2ug23cs102@asmithapes2ug23cs102-VirtualBox:~$
```

Screenshot 2:

```
asmithapes2ug23cs102@asmithapes2ug23cs102-VirtualBox:~$ docker-compose --version
docker-compose version 1.29.2, build unknown
asmithapes2ug23cs102@asmithapes2ug23cs102-VirtualBox:~$
```

Screenshot 3:

```
asmithapes2ug23cs102@asmithapes2ug23cs102-VirtualBox:~/docker-advanced-lab/Docker_lab_advanced$ sudo docker ps
CONTAINER ID   IMAGE                  COMMAND                  CREATED          STATUS          PORTS
                                   NAMES
71ff3a91c015   rabbitmq:3-management   "docker-entrypoint.s…"   16 seconds ago   Up 11 seconds   4369/tcp, 5671-5672/tcp, 15671-1567
2/tcp, 15691-15692/tcp, 25672/tcp   rabbitmq
asmithapes2ug23cs102@asmithapes2ug23cs102-VirtualBox:~/docker-advanced-lab/Docker_lab_advanced$
```

Screenshot 4:

```
asmithapes2ug23cs102@asmithapes2ug23cs102-VirtualBox:~/docker-advanced-lab/Docker_lab_advanced$ docker run -it --name user-a chat-ap
p

--- No Previous Chat History ---

--- Chat Started (unknown-user connecting to localhost) ---
RabbitMQ not ready, retrying... (1/5)
RabbitMQ not ready, retrying... (1/5)
RabbitMQ not ready, retrying... (2/5)
RabbitMQ not ready, retrying... (2/5)
RabbitMQ not ready, retrying... (3/5)
RabbitMQ not ready, retrying... (3/5)
RabbitMQ not ready, retrying... (4/5)
RabbitMQ not ready, retrying... (4/5)
RabbitMQ not ready, retrying... (5/5)
RabbitMQ not ready, retrying... (5/5)

ERROR: Could not connect to RabbitMQ at localhost

Falling back to standalone mode...
==================================================
STANDALONE MODE (No RabbitMQ)
==================================================

--- No Previous Chat History ---

Messages will not be sent to any other container.

Type message (standalone):
ERROR: Could not connect to RabbitMQ at localhost
hello docker
Sent by unknown-user (standalone): hello docker
Type message (standalone):
```

Screenshot 5:

```
asmithapes2ug23cs102@asmithapes2ug23cs102-VirtualBox:~/docker-advanced-lab/Docker_lab_advanced$ docker run -it --name user-a \
> --network chat-net \
> -e RABBbit__HOST=rabbitmq \
> -e QUEUE_NAME=queue_a \
> -e TARGET_QUEUE=queue_b \
> chat-app

--- No Previous Chat History ---

--- Chat Started (unknown-user connecting to localhost) ---
RabbitMQ not ready, retrying... (1/5)
RabbitMQ not ready, retrying... (1/5)
RabbitMQ not ready, retrying... (2/5)
RabbitMQ not ready, retrying... (2/5)
RabbitMQ not ready, retrying... (3/5)
RabbitMQ not ready, retrying... (3/5)
RabbitMQ not ready, retrying... (4/5)
RabbitMQ not ready, retrying... (4/5)
RabbitMQ not ready, retrying... (5/5)
RabbitMQ not ready, retrying... (5/5)

ERROR: Could not connect to RabbitMQ at localhost

Falling back to standalone mode...
==================================================
STANDALONE MODE (No RabbitMQ)
==================================================

--- No Previous Chat History ---

ERROR: Could not connect to RabbitMQ at localhost

Messages will not be sent to any other container.

Type message (standalone): hello docker
Sent by unknown-user (standalone): hello docker
Type message (standalone): what is this
Sent by unknown-user (standalone): what is this
```

Screenshot 6:

```
asmithapes2ug23cs102@asmithapes2ug23cs102-VirtualBox:~/docker-advanced-lab/Docker_lab_advanced$ docker run -it --name user-b --netwo
rk chat-net -e RABBIT_HOST=rabbitmq -e QUEUE_NAME=queue_b -e TARGET_QUEUE=queue_a chat-app

--- No Previous Chat History ---

--- Chat Started (unknown-user connecting to rabbitmq) ---
Type message: hello
Sent by unknown-user: hello
Type message: vm slow:(
Sent by unknown-user: vm slow:(
Type message: what is this
Sent by unknown-user: what is this
Type message:
```

Screenshot 7:

```
asmithapes2ug23cs102@asmithapes2ug23cs102-VirtualBox:~/docker-advanced-lab/Docker_lab_advanced$ docker rm user-a
user-a
asmithapes2ug23cs102@asmithapes2ug23cs102-VirtualBox:~/docker-advanced-lab/Docker_lab_advanced$ docker network rm chat-net
chat-net
asmithapes2ug23cs102@asmithapes2ug23cs102-VirtualBox:~/docker-advanced-lab/Docker_lab_advanced$
```

```
asmithapes2ug23cs102@asmithapes2ug23cs102-VirtualBox:~/docker-advanced-lab/Docker_lab_advanced$ docker stop user-b rabbitmq
user-b
rabbitmq
asmithapes2ug23cs102@asmithapes2ug23cs102-VirtualBox:~/docker-advanced-lab/Docker_lab_advanced$ docker rm user-b rabbitmq
user-b
rabbitmq
```

Screenshot 8:

```
asmithapes2ug23cs102@asmithapes2ug23cs102-VirtualBox:~/docker-advanced-lab/Docker_lab_advanced$ docker-compose up rabbitmq
Creating network "docker_lab_advanced_chat-net" with driver "bridge"
Creating docker_lab_advanced_rabbitmq_1 ... done
Attaching to docker_lab_advanced_rabbitmq_1
rabbitmq_1  | =INFO REPORT==== 12-Feb-2026::10:06:22.107312 ===
rabbitmq_1  |     alarm_handler: {set,{system_memory_high_watermark,[]}}
rabbitmq_1  | 2026-02-12 10:07:11.072206+00:00 [notice] <0.44.0> Application syslog exited with reason: stopped
rabbitmq_1  | 2026-02-12 10:07:11.103259+00:00 [notice] <0.254.0> Logging: switching to configured handler(s); following messages ma
y not be visible in this log output
rabbitmq_1  | 2026-02-12 10:07:11.108534+00:00 [notice] <0.254.0> Logging: configured log handlers are now ACTIVE
rabbitmq_1  | 2026-02-12 10:07:11.305908+00:00 [info] <0.254.0> ra: starting system quorum_queues
rabbitmq_1  | 2026-02-12 10:07:11.306012+00:00 [info] <0.254.0> starting Ra system: quorum_queues in directory: /var/lib/rabbitmq/mn
esia/rabbit@7e8dc9829763/quorum/rabbit@7e8dc9829763
rabbitmq_1  | 2026-02-12 10:07:12.227421+00:00 [info] <0.268.0> ra system 'quorum_queues' running pre init for 0 registered servers
rabbitmq_1  | 2026-02-12 10:07:12.410952+00:00 [info] <0.269.0> ra: meta data store initialised for system quorum_queues. 0 record(s
) recovered
rabbitmq_1  | 2026-02-12 10:07:12.455991+00:00 [notice] <0.274.0> WAL: ra_log_wal init, open tbls: ra_log_open_mem_tables, closed tb
ls: ra_log_closed_mem_tables
rabbitmq_1  | 2026-02-12 10:07:12.647120+00:00 [info] <0.254.0> ra: starting system coordination
rabbitmq_1  | 2026-02-12 10:07:12.647308+00:00 [info] <0.254.0> starting Ra system: coordination in directory: /var/lib/rabbitmq/mne
sia/rabbit@7e8dc9829763/coordination/rabbit@7e8dc9829763
rabbitmq_1  | 2026-02-12 10:07:12.670708+00:00 [info] <0.282.0> ra system 'coordination' running pre init for 0 registered servers
rabbitmq_1  | 2026-02-12 10:07:12.697422+00:00 [info] <0.283.0> ra: meta data store initialised for system coordination. 0 record(s)
 recovered
rabbitmq_1  | 2026-02-12 10:07:12.698300+00:00 [notice] <0.288.0> WAL: ra_coordination_log_wal init, open tbls: ra_coordination_log_
open_mem_tables, closed tbls: ra_coordination_log_closed_mem_tables
rabbitmq_1  | 2026-02-12 10:07:12.770704+00:00 [info] <0.254.0> ra: starting system coordination
rabbitmq_1  | 2026-02-12 10:07:12.770910+00:00 [info] <0.254.0> starting Ra system: coordination in directory: /var/lib/rabbitmq/mne
sia/rabbit@7e8dc9829763/coordination/rabbit@7e8dc9829763
rabbitmq_1  | 2026-02-12 10:07:14.138239+00:00 [info] <0.254.0> Waiting for Khepri leader for 30000 ms, 9 retries left
rabbitmq_1  | 2026-02-12 10:07:14.203968+00:00 [notice] <0.292.0> RabbitMQ metadata store: candidate -> leader in term: 1 machine ve
rsion: 1
rabbitmq_1  | 2026-02-12 10:07:14.254002+00:00 [info] <0.254.0> Khepri leader elected
rabbitmq_1  | 2026-02-12 10:07:14.254200+00:00 [info] <0.254.0> Waiting for Khepri projections for 30000 ms, 9 retries left
rabbitmq_1  | 2026-02-12 10:07:18.445093+00:00 [info] <0.254.0>
rabbitmq_1  | 2026-02-12 10:07:18.445093+00:00 [info] <0.254.0>  Starting RabbitMQ 3.13.7 on Erlang 26.2.5.16 [jit]
rabbitmq_1  | 2026-02-12 10:07:18.445093+00:00 [info] <0.254.0>  Copyright (c) 2007-2024 Broadcom Inc and/or its subsidiaries
rabbitmq_1  | 2026-02-12 10:07:18.445093+00:00 [info] <0.254.0>  Licensed under the MPL 2.0. Website: https://rabbitmq.com
```

Screenshot 9:

```
asmithapes2ug23cs102@asmithapes2ug23cs102-VirtualBox:~/docker-advanced-lab/Docker_lab_advanced$ docker-compose run --rm user-a
Creating docker_lab_advanced_user-a_run ... done

--- No Previous Chat History ---

--- Chat Started (user-a connecting to rabbitmq) ---
Type message: hello
Sent by user-a: hello
Type message: user a what
Sent by user-a: user a what
```

Screenshot 10:

```
asmithapes2ug23cs102@asmithapes2ug23cs102-VirtualBox:~/docker-advanced-lab/Docker_lab_advanced$ docker-compose run --rm user-a
Creating docker_lab_advanced_user-a_run ... done

--- No Previous Chat History ---

--- Chat Started (user-a connecting to rabbitmq) ---
Type message: hello
Sent by user-a: hello
Type message: user a what
Sent by user-a: user a what
```

Screenshot 11:

```
asmithapes2ug23cs102@asmithapes2ug23cs102-VirtualBox:~/docker-advanced-lab/Docker_lab_advanced$ docker-compose run --rm user-a
Creating docker_lab_advanced_user-a_run ... done

--- Chat History ---
Sent by user-a: hello
Sent by user-a: user a what
--- End of History ---

--- Chat Started (user-a connecting to rabbitmq) ---
RabbitMQ not ready, retrying... (1/5)
RabbitMQ not ready, retrying... (1/5)
Type message: chat history
Sent by user-a: chat history
Type message: saved
Sent by user-a: saved
Type message:
```

Screenshot 12:

```
asmithapes2ug23cs102@asmithapes2ug23cs102-VirtualBox:~/docker-advanced-lab/Docker_lab_advanced$ docker-compose run --rm user-b
Building user-b
Step 1/6 : FROM python:3.9-slim
 ---> 085da638e1b8
Step 2/6 : RUN pip install pika
 ---> Using cache
 ---> aec2231ef519
Step 3/6 : WORKDIR /app
 ---> Using cache
 ---> 373ccfcdfcef
Step 4/6 : COPY chat.py .
 ---> Using cache
 ---> ee234a88f1bb
Step 5/6 : RUN mkdir -p /app/data
 ---> Using cache
 ---> e99438f77d63
Step 6/6 : CMD ["python", "-u", "chat.py"]
 ---> Using cache
 ---> 0969d42dd4d2

Successfully built 0969d42dd4d2
Successfully tagged docker_lab_advanced_user-b:latest
WARNING: Image for service user-b was built because it did not already exist. To rebuild this image you must use `docker-compose bui
ld` or `docker-compose up --build`.
Creating docker_lab_advanced_user-b_run ... done

--- Chat History ---
Sent by user-a: hello
Sent by user-a: user a what
--- End of History ---

--- Chat Started (user-b connecting to rabbitmq) ---
Type message:
Received from user-a: chat history
Type message:
Received from user-a: saved
Type message:
```